

MovieHub — Testing Summary

Module: COM661 Full Stack Development

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Application: MovieHub — Full Stack Movie Review Platform

1. Overview

This document covers all automated and manual testing carried out for the MovieHub application across three layers:

- **Frontend unit tests** — Angular services and components (Jasmine / Karma)
 - **Backend integration tests** — Flask API routes (pytest / Flask Test Client)
 - **Manual functional tests** — end-to-end user journeys through the browser
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2. Testing Tools

Layer	Tool	Role
Frontend	Jasmine	Test framework (<i>describe, it, expect</i>)
Frontend	Karma	Test runner — executes specs in Chrome Headless
Frontend	Angular TestBed	Bootstraps Angular dependency injection for unit tests
Frontend	HttpTestingController	Intercepts HTTP calls and asserts URL, method, headers and body
Backend	pytest	Python test runner
Backend	Flask Test Client	Makes HTTP requests directly to the WSGI app

3. How to Run the Tests

Frontend

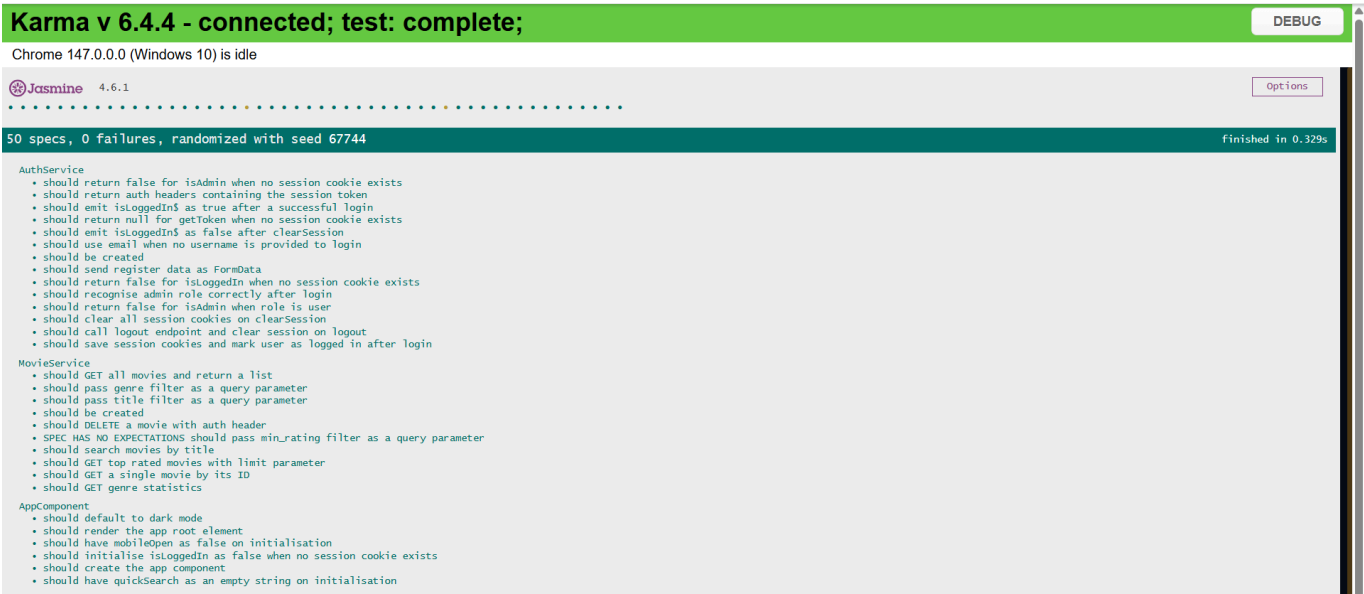
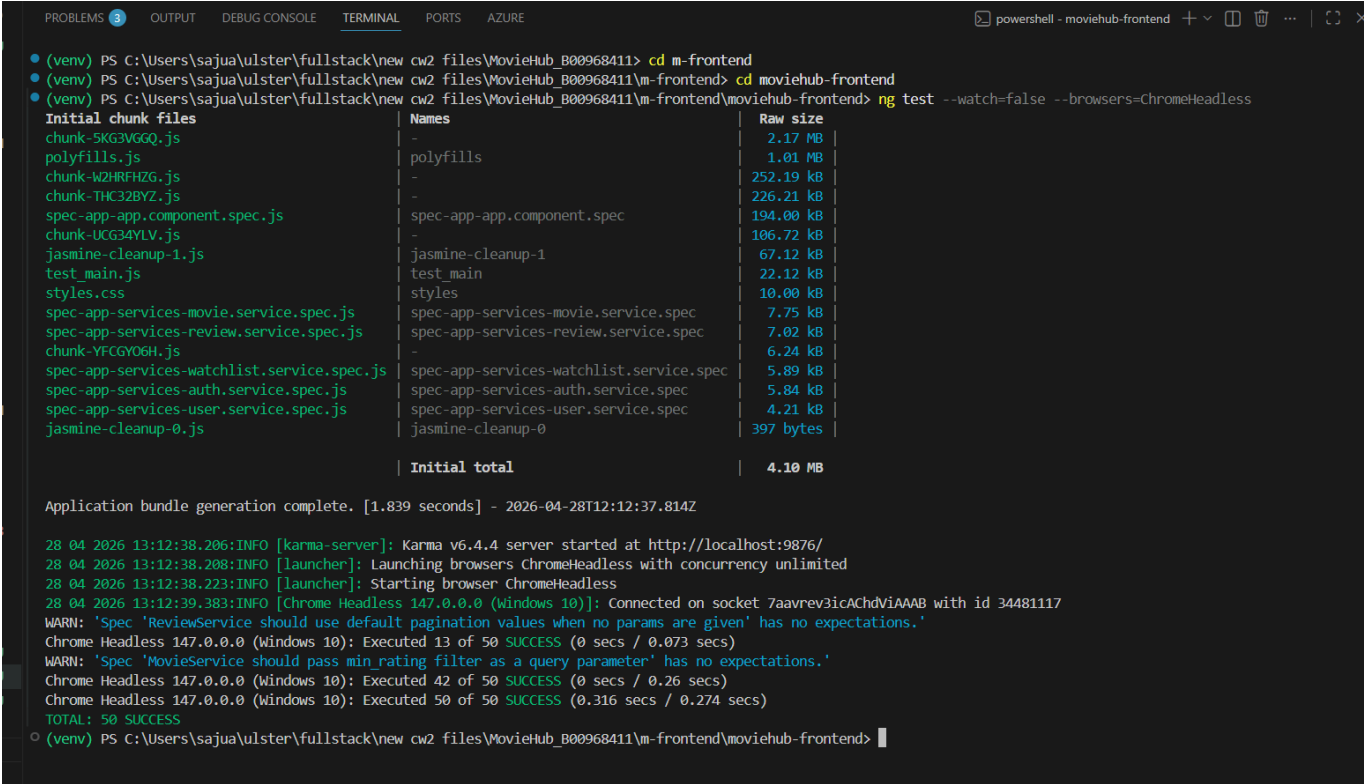
```
cd m-frontend/moviehub-frontend
ng test --watch=false --browsers=ChromeHeadless
```

Backend (requires MongoDB running on *localhost:27017*)

```
cd m-backend
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
python -m pytest test_routes.py -v
```

4. Frontend Test Results

Test Runner: Karma + Jasmine | Browser: Chrome Headless 147 | Result: ✔ 50 / 50 PASSED



Chrome is being controlled by automated test software

```
• should emit isLoggedin$ as false after clearSession
• should use email when no username is provided to login
• should be created
• should send register data as FormData
• should return false for isLoggedin when no session cookie exists
• should recognise admin role correctly after login
• should return false for isAdmin when role is user
• should clear all session cookies on clearSession
• should call logout endpoint and clear session on logout
• should save session cookies and mark user as logged in after login

MovieService
• should GET all movies and return a list
• should pass genre filter as a query parameter
• should pass title filter as a query parameter
• should be created
• should DELETE a movie with auth header
• SPEC HAS NO EXPECTATIONS should pass min_rating filter as a query parameter
• should search movies by title
• should GET top rated movies with limit parameter
• should GET a single movie by its ID
• should GET genre statistics

AppComponent
• should default to dark mode
• should render the app root element
• should have mobileOpen as false on initialisation
• should initialise isLoggedin as false when no session cookie exists
• should create the app component
• should have quickSearch as an empty string on initialisation

ReviewService
• should be created
• should POST a new review as FormData with auth header
• should GET all reviews written by a specific user
• should GET reviews for a movie with correct pagination parameters
• should POST a not_helpful vote with auth header
• SPEC HAS NO EXPECTATIONS should use default pagination values when no params are given
• should POST a helpful vote with auth header
• should DELETE a review with auth header
• should PUT an updated review as FormData with auth header

WatchlistService
• isInWatchlist should return false for any movie when watchlist is empty
• should clear watchlist IDs when user logs out
• watchlistids$ should emit an empty Set initially
• should load watchlist IDs and correctly report which movies are saved
• toggle should navigate to /login when user is not logged in
• should be created

UserService
• should GET all users with auth header
• should GET a single user by ID
• should DELETE a user with auth header
• should PUT a promote request with auth header
• should be created

localhost:9876/context.html?spec=AuthService should return null for getToken when no sessi...
```

4.1 AppComponent — 6 Tests

#	Test	Result
1	should create the app component	✓ PASS
2	should initialise isLoggedin as false when no session cookie exists	✓ PASS
3	should have quickSearch as an empty string on initialisation	✓ PASS
4	should have mobileOpen as false on initialisation	✓ PASS
5	should default to dark mode	✓ PASS
6	should render the app root element	✓ PASS

4.2 AuthService — 14 Tests

#	Test	Result
1	should be created	✓ PASS
2	should return false for isLoggedin when no session cookie exists	✓ PASS
3	should return null for getToken when no session cookie exists	✓ PASS
4	should return false for isAdmin when no session cookie exists	✓ PASS
5	should save session cookies and mark user as logged in after login	✓ PASS
6	should recognise admin role correctly after login	✓ PASS
7	should return false for isAdmin when role is user	✓ PASS

#	Test	Result
8	should clear all session cookies on clearSession	✓ PASS
9	should call logout endpoint and clear session on logout	✓ PASS
10	should return auth headers containing the session token	✓ PASS
11	should emit isLoggedIn\$ as true after a successful login	✓ PASS
12	should emit isLoggedIn\$ as false after clearSession	✓ PASS
13	should use email when no username is provided to login	✓ PASS
14	should send register data as FormData	✓ PASS

4.3 MovieService — 10 Tests

#	Test	Result
1	should be created	✓ PASS
2	should GET all movies and return a list	✓ PASS
3	should pass genre filter as a query parameter	✓ PASS
4	should pass title filter as a query parameter	✓ PASS
5	should pass min_rating filter as a query parameter	✓ PASS
6	should search movies by title	✓ PASS
7	should GET top rated movies with limit parameter	✓ PASS
8	should GET genre statistics	✓ PASS
9	should GET a single movie by its ID	✓ PASS
10	should DELETE a movie with auth header	✓ PASS

4.4 ReviewService — 9 Tests

#	Test	Result
1	should be created	✓ PASS
2	should GET reviews for a movie with correct pagination parameters	✓ PASS
3	should GET all reviews written by a specific user	✓ PASS
4	should POST a new review as FormData with auth header	✓ PASS
5	should PUT an updated review as FormData with auth header	✓ PASS
6	should DELETE a review with auth header	✓ PASS
7	should POST a helpful vote with auth header	✓ PASS

#	Test	Result
8	should POST a not_helpful vote with auth header	✓ PASS
9	should use default pagination values when no params are given	✓ PASS

4.5 UserService — 5 Tests

#	Test	Result
1	should be created	✓ PASS
2	should GET all users with auth header	✓ PASS
3	should GET a single user by ID	✓ PASS
4	should DELETE a user with auth header	✓ PASS
5	should PUT a promote request with auth header	✓ PASS

4.6 WatchlistService — 6 Tests

#	Test	Result
1	should be created	✓ PASS
2	isInWatchlist should return false for any movie when watchlist is empty	✓ PASS
3	should load watchlist IDs and correctly report which movies are saved	✓ PASS
4	toggle should navigate to /login when user is not logged in	✓ PASS
5	watchlistIds\$ should emit an empty Set initially	✓ PASS
6	should clear watchlist IDs when user logs out	✓ PASS

5. Backend Test Results

Test Runner: pytest | **Framework:** Flask Test Client | **Database:** MongoDB localhost:27017

Result: ✓ 46 / 46 PASSED

```
(venv) PS C:\Users\sajua\ulster\fullstack\new_cw2_files\MovieHub_B00968411\m-backend> python -m pytest test_routes.py -v
test_routes.py::TestMovieReadEndpoints::test_get_single_movie_returns_404_for_nonexistent_id PASSED [ 30%]
test_routes.py::TestMovieReadEndpoints::test_get_reviews_for_nonexistent_movie_returns_404 PASSED [ 32%]
test_routes.py::TestMovieWriteEndpoints::test_add_movie_without_auth_returns_403 PASSED [ 34%]
test_routes.py::TestMovieWriteEndpoints::test_update_movie_without_auth_returns_403 PASSED [ 36%]
test_routes.py::TestMovieWriteEndpoints::test_delete_movie_without_auth_returns_403 PASSED [ 39%]
test_routes.py::TestAuthEndpoints::test_register_missing_fields_returns_400 PASSED [ 41%]
test_routes.py::TestAuthEndpoints::test_register_invalid_email_format_returns_400 PASSED [ 43%]
test_routes.py::TestAuthEndpoints::test_register_password_too_short_returns_400 PASSED [ 45%]
test_routes.py::TestAuthEndpoints::test_login_missing_password_returns_400 PASSED [ 47%]
test_routes.py::TestAuthEndpoints::test_login_wrong_credentials_returns_401 PASSED [ 50%]
test_routes.py::TestAuthEndpoints::test_logout_without_token_returns_401 PASSED [ 52%]
test_routes.py::TestAuthEndpoints::test_logout_with_invalid_token_returns_401 PASSED [ 54%]
test_routes.py::TestAuthEndpoints::test_forgot_password_missing_email_returns_400 PASSED [ 56%]
test_routes.py::TestAuthEndpoints::test_forgot_password_unknown_email_still_returns_200 PASSED [ 58%]
test_routes.py::TestAuthEndpoints::test_reset_password_missing_token_returns_400 PASSED [ 60%]
test_routes.py::TestAuthEndpoints::test_reset_password_invalid_token_returns_400 PASSED [ 63%]
test_routes.py::TestAuthEndpoints::test_token_refresh_without_token_returns_401 PASSED [ 65%]
test_routes.py::TestReviewEndpoints::test_add_review_without_auth_returns_401 PASSED [ 67%]
test_routes.py::TestReviewEndpoints::test_edit_review_without_auth_returns_401 PASSED [ 69%]
test_routes.py::TestReviewEndpoints::test_delete_review_without_auth_returns_401 PASSED [ 71%]
test_routes.py::TestReviewEndpoints::test_vote_with_invalid_type_returns_400_before_auth_check PASSED [ 73%]
test_routes.py::TestReviewEndpoints::test_vote_helpful_without_auth_returns_401 PASSED [ 76%]
test_routes.py::TestUserEndpoints::test_get_all_users_without_auth_returns_403 PASSED [ 78%]
test_routes.py::TestUserEndpoints::test_get_user_by_nonexistent_id_returns_404 PASSED [ 80%]
test_routes.py::TestUserEndpoints::test_promote_user_without_auth_returns_403 PASSED [ 82%]
test_routes.py::TestUserEndpoints::test_delete_user_without_token_returns_401 PASSED [ 84%]
test_routes.py::TestWatchlistEndpoints::test_get_watchlist_without_auth_returns_401 PASSED [ 86%]
test_routes.py::TestWatchlistEndpoints::test_get_watchlist_ids_without_auth_returns_401 PASSED [ 89%]
test_routes.py::TestWatchlistEndpoints::test_add_to_watchlist_without_auth_returns_401 PASSED [ 91%]
test_routes.py::TestWatchlistEndpoints::test_remove_from_watchlist_without_auth_returns_401 PASSED [ 93%]
test_routes.py::TestAnalyticsEndpoints::test_analytics_without_auth_returns_403 PASSED [ 95%]
test_routes.py::TestContactEndpoint::test_contact_missing_required_fields_returns_400 PASSED [ 97%]
test_routes.py::TestContactEndpoint::test_contact_valid_submission_returns_200 PASSED [100%]

===== 46 passed in 0.85s =====
(venv) PS C:\Users\sajua\ulster\fullstack\new_cw2_files\MovieHub_B00968411\m-backend>
```

5.1 Home — 2 Tests

#	Endpoint	Expected	Result
1	GET /	200 — welcome message	✓ PASS
2	GET /api/nonexistent-route	404	✓ PASS

5.2 Movie Read Endpoints — 13 Tests

#	Endpoint	Expected	Result
1	GET /api/movies	200 — JSON array	✓ PASS
2	GET /api/movies?genre=Action	200 — filtered list	✓ PASS
3	GET /api/movies?pn=1&ps=3	200 — max 3 items	✓ PASS
4	GET /api/movies?sort_by=title&order=asc	200	✓ PASS
5	GET /api/movies?sort_by=invalid_column	400	✓ PASS
6	GET /api/movies?order=random	400	✓ PASS
7	GET /api/movies/search?title=the	200 — matching results	✓ PASS
8	GET /api/movies/search (no title)	400	✓ PASS
9	GET /api/movies/top-rated	200 — sorted list	✓ PASS
10	GET /api/movies/top-rated?limit=3	200 — max 3 items	✓ PASS
11	GET /api/movies/stats/genres	200 — genre + count objects	✓ PASS

#	Endpoint	Expected	Result
12	GET <code>/api/movies/NONEXISTENT_XYZ_999</code>	404	✓ PASS
13	GET <code>/api/movies/NONEXISTENT_XYZ_999/reviews</code>	404	✓ PASS

5.3 Movie Write Endpoints (Admin Required) — 3 Tests

#	Endpoint	Expected	Result
1	POST <code>/api/movies</code> — no token	403	✓ PASS
2	PUT <code>/api/movies/M001</code> — no token	403	✓ PASS
3	DELETE <code>/api/movies/M001</code> — no token	403	✓ PASS

5.4 Authentication Endpoints — 12 Tests

#	Endpoint	Expected	Result
1	POST <code>/api/auth/register</code> — missing fields	400	✓ PASS
2	POST <code>/api/auth/register</code> — invalid email	400	✓ PASS
3	POST <code>/api/auth/register</code> — password < 6 chars	400	✓ PASS
4	POST <code>/api/auth/login</code> — missing password	400	✓ PASS
5	POST <code>/api/auth/login</code> — wrong credentials	401	✓ PASS
6	POST <code>/api/auth/logout</code> — no token	401	✓ PASS
7	POST <code>/api/auth/logout</code> — invalid token	401	✓ PASS
8	POST <code>/api/auth/forgot-password</code> — missing email	400	✓ PASS
9	POST <code>/api/auth/forgot-password</code> — unknown email	200 (anti-enumeration)	✓ PASS
10	POST <code>/api/auth/reset-password</code> — missing token	400	✓ PASS
11	POST <code>/api/auth/reset-password</code> — invalid token	400	✓ PASS
12	POST <code>/api/auth/refresh</code> — no token	401	✓ PASS

5.5 Review Endpoints — 5 Tests

#	Endpoint	Expected	Result
1	POST <code>/api/movies/M001/reviews</code> — no auth	401	✓ PASS
2	PUT <code>/api/reviews/R001</code> — no auth	401	✓ PASS
3	DELETE <code>/api/reviews/R001</code> — no auth	401	✓ PASS
4	POST <code>/api/reviews/R001/vote/thumbs_up</code> — invalid type	400	✓ PASS

#	Endpoint	Expected	Result
5	POST <code>/api/reviews/R001/vote/helpful</code> — no auth	401	✓ PASS

5.6 User Endpoints — 4 Tests

#	Endpoint	Expected	Result
1	GET <code>/api/users</code> — no admin token	403	✓ PASS
2	GET <code>/api/users/NONEXISTENT_USER_999</code>	404	✓ PASS
3	PUT <code>/api/users/U002/promote</code> — no admin token	403	✓ PASS
4	DELETE <code>/api/users/U001</code> — no token	401	✓ PASS

5.7 Watchlist Endpoints — 4 Tests

#	Endpoint	Expected	Result
1	GET <code>/api/users/U001/watchlist</code> — no auth	401	✓ PASS
2	GET <code>/api/users/U001/watchlist/ids</code> — no auth	401	✓ PASS
3	POST <code>/api/users/U001/watchlist/M001</code> — no auth	401	✓ PASS
4	DELETE <code>/api/users/U001/watchlist/M001</code> — no auth	401	✓ PASS

5.8 Analytics — 1 Test

#	Endpoint	Expected	Result
1	GET <code>/api/admin/analytics</code> — no admin token	403	✓ PASS

5.9 Contact — 2 Tests

#	Endpoint	Expected	Result
1	POST <code>/api/contact</code> — missing fields	400	✓ PASS
2	POST <code>/api/contact</code> — valid submission	200	✓ PASS

6. Manual Functional Testing

6.1 User Registration & Login

#	Action	Expected	Result
1	Submit register form with empty fields	Inline validation errors	✓ PASS
2	Enter invalid email format	Email validation error	✓ PASS

#	Action	Expected	Result
3	Enter password shorter than 6 characters	Strength meter shows Weak	✓ PASS
4	Register with valid credentials	Redirected to home, logged in	✓ PASS
5	Login with wrong password	Invalid credentials error	✓ PASS
6	Login with correct credentials	Username shown in navbar	✓ PASS
7	Attempt 4+ logins in one minute	Rate limit error (429)	✓ PASS

6.2 Session Management

#	Action	Expected	Result
1	Inspect DevTools → Cookies after login	Session cookies visible (<code>moviehub_token</code> etc.)	✓ PASS
2	Inspect localStorage after login	No token in localStorage	✓ PASS
3	Remain active for 5+ minutes	Session maintained (silent refresh)	✓ PASS
4	Logout	Cookies cleared, redirected to home	✓ PASS

6.3 Browse & Filter Movies

#	Action	Expected	Result
1	Navigate to <code>/movies</code>	Paginated movie list loads	✓ PASS
2	Filter by genre "Action"	Only Action movies displayed	✓ PASS
3	Set minimum rating to 8	Only high-rated movies shown	✓ PASS
4	Use quick-search bar	Navigates to filtered results	✓ PASS
5	Sort by release year descending	Newest movies first	✓ PASS

6.4 Movie Reviews

#	Action	Expected	Result
1	View movie detail while logged out	Reviews visible, no review form	✓ PASS
2	Log in and open an unreviewed movie	Review form with 1–10 rating appears	✓ PASS
3	Submit review with rating 8	Review added, movie average updates	✓ PASS
4	Submit a second review for same movie	Error: already reviewed	✓ PASS
5	Edit own review	Rating and comment update correctly	✓ PASS
6	Log in as admin, view movie detail	Review form hidden for admin	✓ PASS
7	Admin deletes another user's review	Review removed, average recalculated	✓ PASS

6.5 Helpful / Not Helpful Votes

#	Action	Expected	Result
1	Click Helpful on another user's review	Count increments	✓ PASS
2	Click Helpful again on same review	Vote removed (toggle)	✓ PASS
3	Click Helpful then Not Helpful	Mutually exclusive — only one active	✓ PASS
4	View own review	Vote buttons not shown	✓ PASS
5	Log in as admin, view reviews	Vote buttons hidden for admin	✓ PASS

6.6 Watchlist

#	Action	Expected	Result
1	Click heart while logged out	Redirected to login	✓ PASS
2	Click heart while logged in	Movie saved, heart fills	✓ PASS
3	Navigate to <code>/watchlist</code>	Saved movies listed	✓ PASS
4	Click heart on a saved movie	Movie removed, heart empties	✓ PASS
5	Reload the page	Watchlist state preserved	✓ PASS

6.7 Password Reset

#	Action	Expected	Result
1	Submit forgot-password with registered email	Success message shown	✓ PASS
2	Check inbox	Reset email received	✓ PASS
3	Click link in email	Reset password page opens	✓ PASS
4	Submit password shorter than 6 characters	Validation error	✓ PASS
5	Submit valid new password	Success, redirected to login	✓ PASS
6	Use same reset link again	Error: invalid or expired link	✓ PASS

6.8 Admin Dashboard

#	Action	Expected	Result
1	Log in as admin, navigate to <code>/admin</code>	Movies tab and Analytics tab visible	✓ PASS
2	Add a new movie	Movie appears in catalogue	✓ PASS
3	Edit an existing movie	Changes saved correctly	✓ PASS
4	Delete a movie	Movie and reviews removed	✓ PASS

#	Action	Expected	Result
5	Click Analytics tab	Four Chart.js charts render	✓ PASS
6	Navigate to <code>/users</code>	All users listed with promote option	✓ PASS

6.9 Dark / Light Mode

#	Action	Expected	Result
1	Click 🌙 in navbar	Theme switches to light	✓ PASS
2	Reload the page	Preference remembered	✓ PASS
3	Click 🌞 in navbar	Theme switches back to dark	✓ PASS

7. Test Coverage Summary

Area	Automated	Manual	Total
AppComponent	6	—	6
AuthService	14	4	18
MovieService	10	5	15
ReviewService	9	7	16
UserService	5	3	8
WatchlistService	6	5	11
Backend Routes	46	—	46
Admin Dashboard	—	6	6
Password Reset	—	6	6
Dark / Light Mode	—	3	3
Total	96	39	135

8. Defects Found & Resolved

#	Defect	Severity	Resolution
1	<code>karma.conf.js</code> referenced a deprecated plugin path incompatible with Angular 21	High	Updated to use the correct <code>@angular/build</code> plugin
2	Auth service tests used <code>localStorage</code> but the service was migrated to cookies — tests failed silently	High	Rewrote test setup to use <code>document.cookie</code> directly

#	Defect	Severity	Resolution
3	Service tests used relative URLs; services use absolute URLs — <code>HttpTestingController</code> never matched	High	Updated all URL strings in spec files to absolute paths
4	<code>RouterTestingModule</code> is deprecated in Angular 21	Low	Replaced with <code>provideRouter([])</code>
5	Review rating displayed as 4/5 but stored as 8/10 — scale mismatch	Medium	Updated all inputs, validation and labels to 1–10 scale
6	<code>expires_at</code> stored as timezone-aware UTC but compared with naive <code>datetime.utcnow()</code> — <code>TypeError</code> on token validation	High	Added <code>.replace(tzinfo=timezone.utc)</code> normalisation in <code>helpers.py</code>